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Laplace Smoothing - Part 1

In the previous lectures, you came across the 'zero probability problem' - the proba never appeared in a class (though it may have appeared in the dataset in another cl

You will now understand how a technique called 'Laplace smoothing' helps solve this p

Note: At 3:13 there is a term $P(\text{cinema})$ missing in the text in the video. though the t numeric calculation.

Note: At 2:01, It should be ' $P(\text{institution/cinema})$ ' instead of ' $P(\text{institution})$ '.

Note: At 1:18, the test document mentions 'Educational' whereas the word is 'educatio

WHY WE NEED LAPLACE SMOOTHING?

Test Document : very good educational institution

↓
ignored (not present in dictionary)

Education

$P(\text{Education} | \text{"good education institution"})$
= $P(\text{good} | \text{Education})$

x $P(\text{educational} | \text{Education})$

	$n_{\text{education}}^{\text{test}}$	$p(w c = \text{education})$	$n_{\text{cinema}}^{\text{test}}$	$p(w c = \text{cinema})$
$w_1 = \text{cinema}$	0	0	1	1/8
$w_2 = \text{depends}$	1	1/13	1	1/8
$w_3 = \text{educational}$	3	3/13	0	0
$w_4 = \text{ethics}$	2	2/13	0	0
$w_5 = \text{good}$	0	0	2	2/8
$w_6 = \text{great}$	2	2/13	1	
$w_7 = \text{greatness}$	2	2/13	0	
$w_8 = \text{institution}$	1	1/13	0	0
$w_9 = \text{movie}$	0	0	1	1/8
$w_{10} = \text{sholey}$	0	0	1	1/8
$w_{11} = \text{story}$	1	1/13	1	1/8
$w_{12} = \text{upgrad}$	1	1/13	0	0

 Study Plan

Laplace Smoothing - Part 1

Wasn't that interesting? Did you notice how we could not assign any class label for o inconclusive results? That is where Laplace smoothing helps. How exactly does it help?

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